

The Right Base

Also Remember...

- Look for onboard USB ports and options for additional USB ports.
- Check for the maximum CPU speed supported.
- Onboard thermal and fan speed monitoring (for servers and power users).
- Voltage and Bus speed tweaking (for the overclockers).
- ATX or BABY AT (ATX power supply has better power management features).
- Check for ACPI compliant BIOS (for better power management).
- Number of IDE devices supported. The higher the better.
- Support for Ultra ATA-100, ATA-66 IDE standards.
- Check for the highest FSB (Front Side Bus) speed supported.
- Support for ECC (Error Correction Capability) RAM.
- Integrated features such as onboard soundcard and display card (for SoHo users).
- Look for features like Wake-on-Modem-ring (integrated into the BIOS) that allows the computer to power on automatically for receiving faxes.

The scenario in the motherboard market has radically changed over the past one year or so. It is a change similar to the Indian automobile market where Indian consumers, who had just a couple of choices earlier, could choose from a glut of cars available after liberalisation. There are now choices for every single type of consumer need, beginning from simple 'Plain Jane' motherboards to fully featured powerhouses for every single type of mainstream processor. However, keeping ahead of the technology curve for motherboards could mean investing thousands of rupees each year. There are ways in which you can avoid this constant upheaval of hardware and it begins by choosing the right technology and feature set. This alone can ensure that your PC is ready to tackle the next-generation hardware. You can now also find a lot of boards and brands offering platforms for the AMD processors.

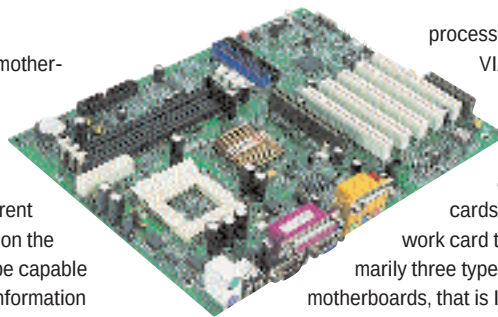
TECHNOLOGY INSIGHT

The primary function of the motherboard is to be able to provide a substrate upon which other components such as the processor and RAM can reside. Since different types of components reside on the motherboard, it also has to be capable of coordinating the flow of information between these various components and orchestrate the transfer of data and information within the system.

SMART TIPS

Processor interface: This is the socket on the motherboard that is used to mount the processor on the board. If categorised according to processor interface, there are generally five types of slots available: Socket 7, Socket 370, Socket A, Slot 1 and Slot A.

Chipset: Different chipsets are used to support functions that are offered by the board, the RAM and, more importantly, the processor used in the system. For example, the Intel i815e chipset supports Pentium III (Coppermine) and Celeron processors up to 1 GHz. AMD



processors (Duron and Athlon) require VIA KT133 chipset based motherboards.

Interface slots: These slots are used to connect add-on cards such as a soundcard or a network card to the computer. There are primarily three types of slots found on today's motherboards, that is ISA (fading away), PCI and AGP. PCI cards are the most common. The AGP interface is meant solely for high-speed graphics cards.

RAM slots: SDRAM is the most popular option although DDR-SDRAM is slowly making headway. Most boards have at least two slots, but more expensive models have four slots for RAM. The total amount of RAM that you can install is also dependant on your motherboard chipset.

BIOS: Every motherboard carries a special circuitry called the BIOS (Basic Input Output System) that exists on a ROM chip. The 'jumperless' motherboards that are the norm today, allow all parameters—ranging from the processor core voltage and clock multiplier to the configuration and control of the integrated peripherals—to be automatically detected and configured.

	Bare necessities	Amateur enthusiast	Performance seeker
You need	Basic support for current processors and peripherals	Flexibility and upgradeability	Maximum performance and overclockability
Look for	Minimum 3 PCI slots with integrated sound and video in the Micro-ATX format	Minimum five PCI slots, three RAM slots, an AGP 4x slot, in the ATX format.	The latest chipset for your processor and options for easy over clocking
Our pick	Mercury KOB815FST (Intel), DFI AM35EC (AMD)	Soltek SL65ME (Intel), Soltek SL75KAV (AMD), AOpen AX4T (Pentium 4)	ASUS CUSL2 (Intel), ASUS A7V133 (AMD), ASUS P4T (Pentium 4)
Price range	Rs 3,000 to Rs 5,000	Rs 5,000 to Rs 8,000	Rs 8,000 and above



The Top Performers on
the Toughest Benchmarks

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